

The swap — somebody else’s data

Name	Group	Date
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You were not there This is the other class’s data. You do not know how they did it. They tested something you did not, and you tested something they did not — neither class can answer the client’s question alone. But before you use their numbers, you have to decide whether you trust them.

WRITE THESE DOWN BEFORE YOU DISCUSS ANYTHING. Once someone says what they think out loud, it gets very hard to notice anything different.

1 • Two things you CANNOT tell from this data

What did they not write down that you would need to know?

2 • One thing you think may have gone wrong

Be specific. Name the step.

3 • One question you would ask them

Things worth looking for Did they dry the nails the same way? Did every jar start on the same day? Were the jars in the same place — same light, same temperature? Same size nails? Is one of their three jars wildly different from the other two, and did they say why? Did they record to two decimal places or round? Is anything missing?

Now turn it around

What would the OTHER CLASS say about OUR method? Name two things they could fairly attack.

1	
2	

Why this matters

9. Could either class have answered the client's question on its own? Explain.
10. Real scientists pull each other's methods apart constantly. Why would that make science BETTER rather than worse?

The point Scientists publish their methods precisely so other people can attack them. A finding that survives other people trying to pull it apart is worth far more than one nobody checked. That is not scientists being difficult with each other — it is the part that makes the answer trustworthy.

Coming up next You are about to write your own investigation plan, and another group is about to do exactly this to it. What would YOU attack?